Linear Transformation

fn. b/w two vector spaces

(here we have it to be $\boxed{V}$ and $\boxed{W}$)

that preserves the operations of

① Vector Addition

② Scalar Multiplication

Eg → If $\boxed{T} : \boxed{V \longrightarrow W}$ from vector space (V) to vector space (W)

Linear Tranformation

then for any vector

2 important properties

if T needs to be called as $\boxed{LT}$ 2 props must be satisfied

① Additivity $T(u+v) = T(u) + T(v)$

② homogenity $T(cv) = c\,T(v)$

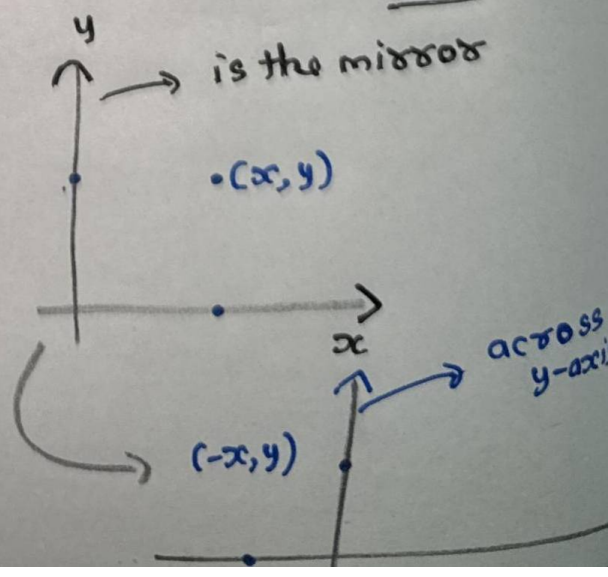
where

$u, v \in \boxed{V}$ vectors in V and $\boxed{c}$ is a scalar value

Eg → Reflection

Reflection Transformation (T) across the y-axis maps a vector

$$x = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \longrightarrow T(x) = \begin{bmatrix} -x \\ y \end{bmatrix}$$



Trans can be expressed as

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{T(x) = AX}$$ DOT PRODUCT

$$= \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

(1×2)

(2×2)

(This has to be same in DOT PRODUCT)

$$\boxed{T(x) = \begin{bmatrix} -x \\ y \end{bmatrix}}$$

① checking Additivity

Let $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$ → 2-D

Real No.

$$T(u+v) = T(u) + T(v)$$

$$u + v = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

$$T(u+v) = A(u+v) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix} = \begin{bmatrix} -(u_1 + v_1) \\ u_2 + v_2 \end{bmatrix}$$

$$T(u) = Au = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -u_1 \\ u_2 \end{bmatrix}$$

$\boxed{L.H.S = R.H.S}$

$$T(v) = Av = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

$$RHS \rightarrow T(u) + T(v) = \begin{bmatrix} -u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -u_1 - v_1 \\ u_2 + v_2 \end{bmatrix}$$

$$\begin{bmatrix} -(u_1 + v_1) \\ u_2 + v_2 \end{bmatrix}$$

② checking homogenity

Homogenity Requirement

$T(cu) = cT(u)$

Let $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in R^2$

and c be a Scalar

$cu = c\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$= \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}$

So,

$T(cu) = A(cu) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}$

$= \begin{bmatrix} -cu_1 \\ cu_2 \end{bmatrix} \rightarrow \boxed{LHS}$

$\boxed{RHS}$

$cT(u) = c(A \cdot u) = c\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$\rightarrow c\begin{bmatrix} -u_1 \\ u_2 \end{bmatrix} \rightarrow \boxed{RHS}$

$\boxed{\text{2nd condition Satisfied}}$

Example that x fallow LT

$$\boxed{\begin{matrix} T: R^2 \rightarrow R^2 \\ u \quad v \end{matrix}}$$

$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\quad T(x) = x + b$

$b \downarrow$ vector (fixed vector)

$$T(x) = x + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

① check Additivity

$T(u+v) = T(u) + T(v)$

So, the first condition itself fails (Both of them shoul pass in order to prove $\boxed{LT}$)

$u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\quad v = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$

$$T(u+v) = T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}\right) = T\left(\begin{bmatrix} 6 \\ 2 \end{bmatrix}\right)$$

$$T\left(\begin{bmatrix} 6 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 7 \\ 3 \end{bmatrix} \rightarrow \boxed{LHS}$$

RHS

$T(u) + T(v)$

$$T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

LHS $\neq$ RHS

$$T\left(\begin{bmatrix} 4 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$T(u) + T(v) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix} \rightarrow RHS$$

② check homogenity

$T(cu) = cT(u)$

$u = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\boxed{c=2}$

$T(cu) = T\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 5 \\ 7 \end{bmatrix} \rightarrow \boxed{LHS}$

$\boxed{RHS} \rightarrow cT(u) = 2\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$

$= 2\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix} \Rightarrow \boxed{RHS}$

$\boxed{LHS \neq RHS}$ × Satisfy homogenity condition

So,

$T(x) = x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow$ is not a linear trans } it fails both additivity homogenity property